

AI for Science: Bridging Machine Learning and Scientific Discovery

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What I cannot create, I do not understand.

Outline

- 1 Introduction: AI for Science
- 2 Physics-Informed Neural Networks (PINNs)
- 3 Molecular Property Prediction
- 4 Climate and Environmental Modeling
- 5 Protein Folding and Structure Prediction
- 6 Materials Discovery
- 7 Scientific Computing and Simulation
- 8 Challenges and Future Directions
- 9 Selected papers
 - AlphaFold 3
 - UniGenX
 - RFDiffusion
 - MPNN
- 10 Conclusion

What is AI for Science?

AI for Science represents the intersection of artificial intelligence and scientific research, where machine learning methods are applied to accelerate scientific discovery.

- **Drug Discovery**: Predicting molecular properties and designing new compounds
- **Materials Science**: Discovering new materials with desired properties
- **Climate Modeling**: Understanding and predicting climate patterns
- **Physics**: Solving complex physical systems and equations
- **Biology**: Understanding protein folding and biological processes

Key Challenges in AI for Science

Scientific problems present unique challenges for machine learning:

- ① **Data scarcity:** Limited experimental data, expensive to collect
- ② **Physical constraints:** Solutions must obey physical laws
- ③ **Interpretability:** Scientific understanding requires explainable models
- ④ **Uncertainty quantification:** Scientific predictions need confidence bounds
- ⑤ **Multi-scale phenomena:** From quantum to macroscopic scales

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Physics-Informed Neural Networks

PINNs incorporate physical laws directly into neural network training:

$$\mathcal{L} = \mathcal{L}_{data} + \lambda \mathcal{L}_{physics} \quad (1)$$

where

- $\mathcal{L}_{data}$: Standard data fitting loss
- $\mathcal{L}_{physics}$: Physics constraint loss (PDEs, conservation laws)
- λ : Weighting parameter balancing data and physics

Example: Solving PDEs with PINNs

Consider the heat equation:

$$\frac{\partial u}{\partial t} = \alpha \nabla^2 u + f(x, t) \quad (2)$$

with boundary conditions $u(x, 0) = g(x)$ and $u(\partial\Omega, t) = h(x, t)$.

The physics loss becomes:

$$\mathcal{L}_{physics} = \left\| \frac{\partial u_{NN}}{\partial t} - \alpha \nabla^2 u_{NN} - f \right\|^2 + \lambda_1 \|u_{NN}(x, 0) - g(x)\|^2 + \lambda_2 \|u_{NN}(\partial\Omega, t) - h(x, t)\|^2 \quad (3)$$

where $u_{NN}(x, t; \theta)$ is the neural network approximation.

Advantages of PINNs

- **No mesh required:** Unlike traditional numerical methods
- **Unified framework:** Handle forward and inverse problems
- **Data fusion:** Combine sparse data with physical knowledge
- **Parameter discovery:** Learn unknown parameters in PDEs

Challenges:

- Training can be unstable
- Requires careful hyperparameter tuning
- May struggle with complex geometries

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Molecular Representation Learning

Converting molecules to machine-readable formats:

- **SMILES strings:** Text-based molecular representation
- **Graph neural networks:** Molecular graphs with atoms as nodes, bonds as edges
- **3D coordinates:** Spatial arrangement of atoms
- **Descriptors:** Hand-crafted molecular features

Graph Neural Networks for Molecules

For a molecular graph $G = (V, E)$ with atoms $v_i \in V$ and bonds $e_{ij} \in E$:

Message passing:

$$h_v^{(l+1)} = \text{UPDATE} \left(h_v^{(l)}, \text{AGGREGATE} \left(\{h_u^{(l)} : u \in \mathcal{N}(v)\} \right) \right) \quad (4)$$

Property prediction:

$$y = \text{READOUT}(\{h_v^{(L)} : v \in V\}) \quad (5)$$

where L is the number of message passing layers.

Applications in Drug Discovery

- **Property prediction:** Solubility, toxicity, binding affinity
- **Molecular generation:** Designing new drug candidates
- **Retrosynthesis:** Planning synthetic routes
- **Protein-ligand binding:** Predicting drug-target interactions

Key datasets:

- QM9: 134k small organic molecules; GEOM-Drugs
- ChEMBL: Large-scale bioactivity data
- PDB: Protein structures; AlphaFoldDB

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Climate systems involve complex interactions across multiple scales:

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} = -\frac{1}{\rho} \nabla p + \nu \nabla^2 \mathbf{u} + \mathbf{F} \quad (6)$$

where $\mathbf{u}$ is velocity, p is pressure, and $\mathbf{F}$ represents external forces.

Traditional approach: Numerical weather prediction (NWP) using physics-based models.

AI approach: Deep learning models trained on historical weather data:

- **ConvLSTM:** Spatiotemporal weather prediction
- **Graph neural networks:** Modeling atmospheric dynamics
- **Transformer models:** Long-range dependencies in weather patterns; ViT; Swin Transformer;

Hybrid approach: Combining AI with traditional NWP for improved accuracy.

Climate Change Detection

Using AI to detect and attribute climate change:

- **Pattern recognition:** Identifying climate change signatures
- **Attribution studies:** Linking extreme events to climate change
- **Scenario modeling:** Predicting future climate under different emissions

Key challenges:

- Limited historical data
- Non-stationary climate system
- Uncertainty quantification

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The Protein Folding Problem

Proteins fold from linear amino acid sequences to 3D structures:

Sequence: ACDEFGHIKLMNPQRSTVWY $\rightarrow$ 3D Structure (7)

Key challenges:

- Exponential number of possible conformations
- Complex energy landscapes
- Multiple timescales (nanoseconds to seconds)

AlphaFold and Structure Prediction

AlphaFold approach:

- ① **Multiple sequence alignment:** Evolutionary information
- ② **Attention mechanisms:** Long-range interactions
- ③ **Structure module:** 3D coordinate prediction
- ④ **Confidence estimation:** Reliability of predictions

Key innovations:

- End-to-end differentiable architecture
- Integration of evolutionary and physical constraints
- Uncertainty quantification

Impact of AlphaFold

- **Structural biology revolution:** Millions of protein structures predicted
- **Drug discovery:** Better understanding of drug targets
- **Protein design:** Engineering proteins with desired functions
- **Evolutionary biology:** Understanding protein evolution

Current limitations:

- Limited accuracy for disordered regions
- Challenges with protein complexes
- Difficulty with membrane proteins

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Predicting material properties from composition and structure:

- **Electronic properties:** Band gap, conductivity, magnetic properties
- **Mechanical properties:** Elastic modulus, strength, hardness
- **Thermal properties:** Heat capacity, thermal conductivity
- **Optical properties:** Refractive index, absorption spectra

Crystal Structure Representation

Periodic systems: Representing crystal structures with periodic boundary conditions.

Graph representations:

- **Crystal graphs:** Atoms as nodes, bonds as edges
- **Periodic embeddings:** Handling translational symmetry
- **Multi-scale features:** From atomic to macroscopic properties

Deep learning approaches:

- **CGCNN:** Crystal Graph Convolutional Neural Networks
- **SchNet:** Quantum-chemical property prediction
- **MEGNet:** Materials property prediction

Computational screening:

- ① Generate candidate materials
- ② Predict properties using ML models
- ③ Filter promising candidates
- ④ Experimental validation

Key databases:

- **Materials Project**: 150k+ materials with computed properties
- **OQMD**: Open Quantum Materials Database
- **AFLOW**: Automatic Flow for Materials Discovery

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Accelerating Scientific Simulations

AI can accelerate traditional scientific computing:

- **Surrogate models:** Fast approximations of expensive simulations, fast and efficient such as in nuclear fusion
- **Multiscale modeling:** Bridging different length and time scales
- **Parameter optimization:** Finding optimal simulation parameters
- **Uncertainty quantification:** Propagating uncertainties through simulations

Neural Operators

Learning mappings between function spaces:

$$G : \mathcal{U} \rightarrow \mathcal{V}, \quad u \mapsto v \quad (8)$$

where $\mathcal{U}$ and $\mathcal{V}$ are function spaces.

Examples:

- **Neural ODEs:** Learning dynamics of ordinary differential equations
- **Neural PDEs:** Solving partial differential equations
- **Fourier Neural Operators:** Learning in frequency domain

- **Fluid dynamics:** Turbulence modeling, flow prediction
- **Quantum chemistry:** Electronic structure calculations
- **Solid mechanics:** Stress analysis, fracture prediction
- **Electromagnetics:** Wave propagation, antenna design

Benefits:

- Orders of magnitude speedup
- Reduced computational cost
- Real-time predictions

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Current Challenges in AI for Science

- ① **Data quality and quantity:** Limited, noisy, or biased scientific data
- ② **Physical consistency:** Ensuring AI predictions obey physical laws
- ③ **Interpretability:** Understanding what AI models learn
- ④ **Generalization:** Models that work across different domains
- ⑤ **Uncertainty quantification:** Reliable error estimates

- **Foundation models for science:** Large-scale models trained on scientific data
- **Automated hypothesis generation:** AI systems that propose new scientific theories
- **Scientific reasoning:** AI that can perform logical scientific reasoning
- **Multimodal scientific AI:** Integrating different types of scientific data
- **Scientific collaboration:** AI systems that work alongside human scientists

- **Scientific integrity:** Ensuring AI doesn't introduce bias in scientific research
- **Reproducibility:** Making AI for science research reproducible
- **Open science:** Sharing models, data, and code
- **Scientific communication:** How AI affects scientific publishing and peer review

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AI for Science represents a paradigm shift in how we approach scientific research:

- **Acceleration:** Faster discovery and development cycles
- **Integration:** Combining data-driven and physics-based approaches
- **Automation:** Reducing manual work in scientific research
- **Insights:** Discovering patterns humans might miss

The Future of AI for Science

- **Democratization:** Making advanced scientific tools accessible to all researchers
- **Interdisciplinary collaboration:** Breaking down silos between scientific fields
- **Scientific method evolution:** How AI changes the scientific process
- **Grand challenges:** Tackling humanity's biggest scientific problems

Questions for Discussion

- ① How can we ensure AI for Science maintains scientific rigor and reproducibility?
- ② What are the implications of AI for the future of scientific careers?
- ③ How should we balance AI automation with human scientific intuition?
- ④ What ethical guidelines should govern AI for Science research?

Welcome inputs

Any comments?

Welcome your inputs!